Code No: 114CS **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech II Year II Semester Examinations, April/May - 2023
DESIGN AND ANALYSIS OF ALGORITHMS
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART - A

(25 Marks)

1.a)	Define Theta notation.	[2]
b)	Write the binary search worst case analysis with example.	[3]
c)	Define binary search tree.	[2]
d)	Write union and find algorithms.	[3]
e)	What is the advantage of dynamic programming?	[2]
f)	Write the application area of Greedy strategy.	[3]
g)	List the differences between fractional and 0/1 knapsack problems.	[2]
h)	Define Hamiltonian cycle.	[3]
i)	Define the term linear time complexity.	[2]
j)	Is Chromatic number decision problem is NP-Hard? Justify.	[3]

PART - B

(50 Marks)

2.a) Explain the concept of Amortized complexity.
b) Find Big-Oh Notation and Omega Notation for $f(n)=7n^3+50n^2+200$. [5+5]

OR

3.a) Write the algorithms to merge two sorted list and analyze its worst case time complexity.
b) Evaluate the 2 × 2 dimensional matrix multiplication time complexity. [5+5]

4. Write a non-recursive algorithm for binary tree traversals. [10]

OR

5.a) Write the algorithm for breadth first search graph traversal strategy.
b) Compare and contrast connected components and bi connected components. [5+5]

6. Explain and write a greedy algorithm for Job Sequencing with Deadlines. [10]

OR

7. Explain the minimum cost spanning tree prim's strategy. Write down algorithm for it and derive its time complexity. [10]

8.a) Interpret the method of reduction to solve TSP problem using Branch and Bound.
b) Explain 8-Quenes problem. [5+5]

OR

9. Give LCBB solution for the following Knapsack instance n=4, (p1,p2,p3,p4)=(10,10,12,18), (w1,w2,w3,w4)=(2,4,6,9) & m=15. [10]

10. Compare and contrast deterministic and non-deterministic algorithms. [10]

OR

11. Explain the strategy to prove that a problem is NP-complete. [10]

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